
The Price of a Flat Direction: Transverse Curvature Sets Retention in a Trained Recurrent Memory

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Abstract

1 A recurrent state can store a continuous value on a manifold of marginally stable
2 states, and this literature already knows that such a manifold is fragile: noise
3 diffuses the state along it, most infinitesimal perturbations of the dynamical law
4 destroy it, and corrective mechanisms that keep it alive are classic. *What has not*
5 *been available is the exchange rate.* We measure one, on a trained potential. The
6 object is a recurrent unit advancing a latent state (q, p) by a damped symplectic
7 velocity-Verlet step of a learned Hamiltonian, whose trained potential V_θ is G -
8 invariant with minimisers on a G -orbit, so that the curvature μ transverse to that
9 orbit prices how long a written value survives. The price list has three parts, every
10 number at latent dimension 4 on an S^1 testbed, ≤ 5 seeds, laptop CPU. Retention
11 half-life scales as $n_{1/2} \propto \mu^{-2}$, at fitted overdamped slope -0.985 against a
12 predicted -1 , the underlying curvature identity exact to $1.000000 \pm 5 \times 10^{-12}$
13 over 4.5 decades. The law stops at a crossover $\varepsilon\mu \approx \gamma/2$, past which the half-
14 life saturates at a curvature-independent floor. And the whole list survives the
15 training-time correction that keeps the orbit alive, at $\approx 20\times$ the horizon on which
16 the objective would otherwise destroy it. Its evidence on a learned system is a
17 head-to-head: a recently posted machine-learning lifetime law, run unchanged
18 on our trained checkpoints, is this price list's overdamped face — matching to
19 $\leq 3\%$ below the crossover and mispredicting by $\approx 3.2\times$ above it, in the calculable
20 direction. *We therefore also report where the price list does not extend, and those*
21 *boundaries are part of the result:* on this architecture class symmetric training
22 data alone does not produce a near-flat direction, a gap of 13–14 orders in μ^2
23 that is a measurement on our recipe and never a general claim about learning; the
24 continuous coset register is designed-only; the raw retention ratio against recurrent
25 baselines inverts under compute normalization and is retired as a compute claim;
26 and four positioning claims are retired against prior art. Designed-testbed results
27 are labeled verification; learned-system results, evidence.

28 1 Introduction

29 A recurrent state holding an analog value can be built on a flat direction: a set of states the dynamics
30 do not distinguish, so a written value neither decays nor is pulled to a preferred point. Symmetry
31 supplies such directions for free. If a learned potential V_θ is G -invariant and its minimiser is not, the
32 minimisers form a G -orbit, the Hessian annihilates every orbit tangent, and the orbit coordinate is
33 neutral. *However*, a flat direction is only as useful as its price list, and none existed for a trained model:
34 how much retention a given transverse curvature buys, where that law stops holding, and whether it
35 survives the correction that keeps the orbit alive. *We measure all three.* The object advances (q, p) by
36 a symplectic velocity-Verlet step of a learnable separable Hamiltonian $H(q, p) = T(p) + V_\theta(q)$, with

37 a per-step damping $p \mapsto (1 - \gamma)p$ supplying controllable forgetting. This is not a new architecture
 38 but a closed-form account of retention on a trained one. The reference unit is the CLU (Causal
 39 Learning Unit), introduced as CHLU in Jawahar & Pierini (2026); results hold for the class of damped
 40 symplectic recurrences, and the exactly-solvable theory verified here is developed in a separate note,
 41 cited as *the theory note* (Anonymous, 2026, in the supplementary material).

42 **Two masses, and one conversion.** Two quantities here are called “mass” and run in opposite
 43 directions. The inertial mass M_i is the learned diagonal of T : larger M , slower. The spectral mass
 44 μ_k , which we lead with as the *transverse curvature*, is $\mu_k^2 = \lambda_k(M^{-1/2}\nabla^2 V_\theta M^{-1/2})$ at a critical
 45 point: larger μ , shorter memory. All retention statements use μ . One conversion precedes any
 46 cross-field comparison: flow-Jacobian eigenvalues are inverse time, whereas μ^2 are eigenvalues of
 47 a mass-whitened Hessian of a *potential*, inverse time squared. Lifetimes are half-lives $n_{1/2}$ in map
 48 applications, not τ and not diffusion coefficients D ; each is reported with the metric defining it,
 49 because those metrics bifurcate past the crossover.

50 **Contribution.** This paper makes one, and it is quantitative: *the transverse-curvature price list of*
 51 *a trained recurrent memory* (§4.1), in three parts — what a given curvature buys, where the law
 52 stops, and whether it survives the training-time correction. *Its evidence on a learned system* is the
 53 head-to-head of §4.2: a recently posted lifetime law, run unchanged on our trained checkpoints, is
 54 the price list’s overdamped face and fails above the crossover in the direction the list predicts. *Its*
 55 *honest negative* is the boundary of §4.3: on this architecture class the flat direction has to be designed
 56 rather than trained for, the continuous register does not transfer to an emergent potential, and the
 57 retention advantage over recurrent baselines inverts under compute normalization. Designed-tested
 58 results — invariant potentials, analytic tilts and spurions — are verification of the theory’s exactness;
 59 learned-system, training-dynamics and head-to-head results are evidence.

60 2 Related work

61 Two questions this audience is asking — how neural representations behave *dynamically*, and what
 62 symmetry buys a learning system — meet in one object, and this paper is about the price of that object.
 63 Neutrality of group-orbit directions is classical (Golubitsky, Stewart & Schaeffer 1988; Krupa 1990;
 64 Rumberger 2001). The same object is studied as a *continuous attractor*: a manifold whose points
 65 are all fixed points, “marginally stable tangent to the manifold and stable normal to the manifold”
 66 (Ságodi et al. 2024), the substrate of the neural integrator (Seung 1996) and of measured ring and
 67 toroidal codes (Kim et al. 2017; Gardner et al. 2022; Khona & Fiete 2022). We make no claim about
 68 biological systems and do not model neural data.

69 *Five facts about that object are established, and none of them is ours.* Noise diffuses the coordinate
 70 along the manifold, the diffusion coefficient lower-bounded by the code’s own Fisher information
 71 (Burak & Fiete 2012). The manifold is structurally unstable, destroyed by most infinitesimal changes
 72 of the dynamical law, with online learning named as one perturbation source — the *fine-tuning*
 73 *problem*, whose memory error Ságodi et al. bound by the unknown perturbation’s size, worst-case
 74 and linear in t . A homeostatic mechanism that restores a destroyed flat direction is 23 years old
 75 (Renart, Song & Wang 2003), so a corrective term is not our idea, and nothing here is a first report
 76 of the destroy-and-restore pattern. A local learning rule can *produce* accurate ring-attractor tuning
 77 (Vafidis et al. 2022), so nothing we measure implies that learning cannot build a flat direction. And
 78 approximate continuous attractors arise in other task-trained recurrent networks by themselves, with
 79 a recent analytical model of the slow structure appearing during training (Dinc et al. 2025, preprint),
 80 where §4.4 runs the opposite direction. *Absent is the exchange rate: how much retention a given*
 81 *transverse curvature buys, where that law stops holding, and whether it survives the correction. We*
 82 *measure all three on trained models, in closed form, with a regime map — floor, crossover — that a*
 83 *first-order flow cannot exhibit.*

84 Closest to our object among this audience’s own work is Xu et al. (2023), where a Lie-group
 85 representation is carried in the recurrent state of a grid-cell network and the compact, connected,
 86 abelian representation forces the torus topology over a continuous attractor. That is our geometry, in a
 87 network that path-integrates; what we add to it is a price list for time. Two further neighbours ask the
 88 same question in different currency: the optimal packing of attractor states in a representation, where
 89 symmetries of the environment’s transition statistics imply symmetries of the optimal code (Vastola

2024), and memory modification approached through symmetry and geometry (Dönmez 2024). Exact-versus-relaxed equivariance is this audience’s own name for our designed-versus-emergent axis (van der Ouderaa & van der Wilk 2023). Reading a landscape by its level sets and critical points is likewise native here (Akhtiamov & Thomson 2023; Wang & Ponce 2023, whose level sets are of tuning landscapes over *stimuli*, not of a latent-state potential — the two must not be conflated). A quotient coordinate has been used as an input encoding (Aslan, Platt & Sheard 2023), where ours is a storage device.

Mo (2026) proves that a C^1 field exactly equivariant under a Lie group has at least $\dim(G/\mathcal{H})$ zero Lyapunov exponents tangent to the orbit; we use his diagnostics and his breaking-and-censoring protocol, and §4.2 runs his estimator unchanged on our trained models. His result is the *kinematics* of protection, silent on what sets the gap; here the gap is *constitutive*, and Lyapunov spectra are blind to the write semantics — transport, latch, noise accumulation — that make a neutral direction a *memory*. His theorem states sufficiency; equivariance is not necessary, the latch being a modulus of V_θ . Constructive equivariant memory (Di Bernardo et al. 2025; Keller 2025; Iqbal, Keller, Song, Miyato & Welling 2026) engineers *which* symmetry a network expresses but does not price *time*: we claim no novelty for choosing a symmetry group or coset, only the price list attached to a chosen structure. Cross-session representational drift (Driscoll et al. 2017; Aitken et al. 2022; Jude et al. 2023) is a different phenomenon from motion along a flat direction and we do not address it; the only motion we report is of the coset coordinate under our own dynamics. Our comparators are LSTM (Hochreiter & Schmidhuber 1997), LEM (Rusch et al. 2022) and coRNN (Rusch & Mishra 2021a); conformal symplecticity is McLachlan & Perlmutter (2001), the leapfrog $h < 2$ limit standard (Hairer, Lubich & Wanner 2006), and wake–sleep (Hinton et al. 1995) with PCD (Tieleman 2008) the contrastive objective, whose landscape distortion is classical (Fischer & Igel 2010; Nijkamp et al. 2020).

Four claims we retire, so that a reviewer’s first objections are answered in writing (elaborated, with their numbers, in Appendix E). (1) Attention and state-space models *do* have retention-guarantee analogues, and HiPPO-LegS (Gu et al. 2020) is the counterexample. (2) “Retrieval is a rollout, not a weighted sum” is Kong, Brewer & Lai’s (2024) move, not our novelty. (3) Continuity is no escape hatch from the NTM/DNC lineage (Graves et al. 2014, 2016; Csordás & Schmidhuber 2019), which was already fully soft. (4) This is no competitor to the transformer, and was never claimed to be. Our claim is correspondingly narrow: a physics-structured associative memory in which retention is a designed, per-item property — permanent (symmetry-protected, $\mu^2 \equiv 0$ exactly) or of a finite half-life set by μ^2 — rather than an emergent consequence of a learned recurrence — *controllability*, which we evidence, not *capability*, which we do not claim against transformers or state-space models. The nearest published neighbour to our write is Titans (Behrouz et al. 2025), test-time gradient descent with momentum on an associative-memory loss: the same family of discretized damped second-order dynamics, differing in that our damping and inertia are *read out and priced*, not optimizer hyperparameters.

3 Setup

The map, and the three bands. One dissipative velocity-Verlet step with step ε and damping γ advances (q, p) by a kick–drift–kick of $H = T(p) + V_\theta(q)$ followed by $p \mapsto (1 - \gamma)p$. It is conformally symplectic — $J^\top \Omega J = (1 - \gamma)\Omega$, $\det J = (1 - \gamma)^d$: phase-space volume contracts by a fixed factor per step, so the geometry is preserved and only the scale is spent. Memory at each normal mode reduces to one 2×2 block ($\mu^2 = k/m$, $h = \varepsilon\mu$) with three bands: a latch ($\mu = 0$, symmetry-protected) with frozen displacement $q_\infty = q_0 + \varepsilon p_0/(M\gamma)$ and infinite half-life; an overdamped register ($0 < \varepsilon\mu \lesssim \gamma/2$) at $n_{1/2} \approx 2\gamma \ln 2/[(2 - \gamma)(\varepsilon\mu)^2]$; and an underdamped working memory ($\gamma/2 \lesssim \varepsilon\mu < 2$) at the curvature-independent envelope half-life $2 \ln 2/(-\ln(1 - \gamma))$. The boundary is the exceptional point $h^* \approx \gamma/2$, the overdamped-to-underdamped crossover at which the block becomes defective. Relaxation onto the orbit is the fast normal flow; retention is the slow tangent flow. One difference from a classical neural integrator we do not smooth over: an integrator stores by *not* damping, whereas at $\gamma = 0$ our flat mode never freezes a write and any $\gamma > 0$ gives an exact latch.

What the curvature can be, and what the unit stores. At a critical point of V_θ the transverse curvature takes one of three values, and trained models here realize all three: *exactly* zero along $\dim(G/\mathcal{H})$ directions when the trained minimisers form a G -orbit (a Nambu–Goldstone mode — the designed unit, and the exactly-flat-direction geometry); small but nonzero when an explicit tilt breaks

the orbit, giving a slow manifold rather than a flat one (a pseudo-Goldstone mode — the *explicitly-broken cell*, which the analytic tilts of Appendix F instantiate and the self-broken MLP exhibits); and strictly positive with no orbit of minimisers at all when the minimiser is itself invariant, degenerate within each irrep by Schur’s lemma (the contrastive-divergence–eroded vacuum of §4.4). Only the *classical, tree-level* Goldstone theorem is used, a statement about a Hessian. No thermodynamic limit, no phase transition, no $\hbar$, and no loops, chiral logarithms, running low-energy constants or anomalies anywhere in this paper. What the unit stores is the *coset coordinate*, the position along the flat direction; the map’s (ε, γ, T) fixes what a given μ^2 buys, through the three bands above.

Trained models. All checkpoints are the reference unit’s Experiment-D configuration — dim 4, hidden 64, learned-Newtonian kinetic term, $dt = 0.05$, unit circle at 256 points, 150 epochs (§4.4 for why not the default 1000) — in two families: designed, an architecturally- $SO(2)$ -invariant V_θ on 5 seeds, and emergent, a generic MLP V_θ on the same data on 3 seeds.

Fine print, next to the claims it qualifies. (a) The latch’s infinite half-life is a $T = 0$ statement about the deterministic damped map; at $T > 0$ the coset coordinate diffuses with $D_\theta = \varepsilon T(2 - \gamma)/(2F^2\gamma)$, $F^2 = M_{\text{ch}} r^{*2}$ — a *diffusive register* of finite computable lifetime, verified on the trained designed checkpoints, where raising friction *lengthens* coset memory by $3.77 \pm 0.23\times$ for $\gamma : 0.05 \rightarrow 0.2$ on 5/5 seeds, and holding only under fluctuation–dissipation-consistent noise and a Newtonian kinetic mode, never under the reference implementation’s legacy noise default. All §4 results are $T = 0$. (b) By Sylvester’s law of inertia a flat direction keeps $\mu^2 = 0$ under any inertial anisotropy: symmetry protects the write *current*, flatness protects the *register*, and map equivariance is sufficient but not necessary for a neutral memory direction — our kinetically-broken battery is a *measured* non-equivariant map that latches exactly (Appendix D). (c) The *tilt-as-a-designed-lifetime-dial* construction is claimed for designed geometries only, the self-broken MLP being an *observed* instance of the cell and not a dial: in a separate battery of ours on a *learned, multi-atom* store, a breaking coefficient intended as the lifetime dial moved the softest eigenvalue *the wrong way*, and a coercivity term — not the breaking — set the ceiling (Appendix D).

Mandatory flag, travels with every finite-temperature number in this paper. All such results require `langevin_noise="fddt"` — the fluctuation–dissipation-consistent noise scale $\sigma_i^* = \sqrt{M_i T \gamma (2 - \gamma)}$ — and a Newtonian kinetic mode (`newtonian_identity/newtonian_learned`). The reference implementation’s default is `legacy`, under which T is not in energy units and none of these laws hold; and in the relativistic kinetic mode no noise scale targets Gibbs at all (Appendix D; *theory note*, Prop-9’), so the laws are undefined there. The checkpoints here are `newtonian_learned`, so every finite-temperature number here is in scope. Nothing in §4 depends on it: all main-text retention results are $T = 0$ deterministic rollouts.

4 Results: the price list, its evidence, and its boundary

4.1 What a curvature buys, and where the law stops

A zero eigenvalue certifies that a direction is neutral. *It does not say what a nearly-zero one costs. We therefore tilt a trained potential by a known amount, read the resulting transverse curvature off its Hessian, and measure the half-life of a written value.* On the trained designed vacuum, whose flat mode has $\mu^2 \leq 2.4 \times 10^{-15}$ and is machine-flat, the constitutive identity holds: the Hessian-derived spectral mass equals the measured one-step-Jacobian gap to 3.2×10^{-10} over all 70 tilt rows. The retention law follows, with the curvature ratio $\mu_{\text{meas}}^2 / \mu_{\text{pred}}^2 = 1.000000 \pm 5 \times 10^{-12}$ at every tilt and seed across 4.5 decades and a fitted overdamped slope of -0.985 ($n = 35$; predicted -1). Past the crossover the half-life stops depending on curvature at all, saturating at the floor 27.03 steps at $\gamma = 0.05$. Two second-order signatures confirm that the crossover is a genuine exceptional — defective — point and not a fitting artifact: the retention metric bifurcates there, envelope half-life and first-crossing time agreeing to 0.01% overdamped and splitting by $3.2\times$ at the deepest tilt, which is why every lifetime in this paper names its metric; and the oscillation frequency turns on with the predicted square root, $\varphi = 0$ exactly below h^* (15/15 rows) and $\varphi \propto \sqrt{h - h^*}$ above it at fitted slope 0.5165 against the predicted 1/2. The latch corner of the same block is the transport law $q_\infty = q_0 + \varepsilon p_0 / (M\gamma)$, which holds to $\leq 1\%$ over two decades of γ , the written angle then frozen to $\leq 1.2 \times 10^{-15}$ rad. *These are trained checkpoints carrying analytic tilts: verification of the theory’s exactness on a learned potential, not a discovery.* Appendix F records the battery in full, the probe’s own roundoff floor, and the damping corollary that makes $n_{1/2}(\gamma)$ non-monotone.

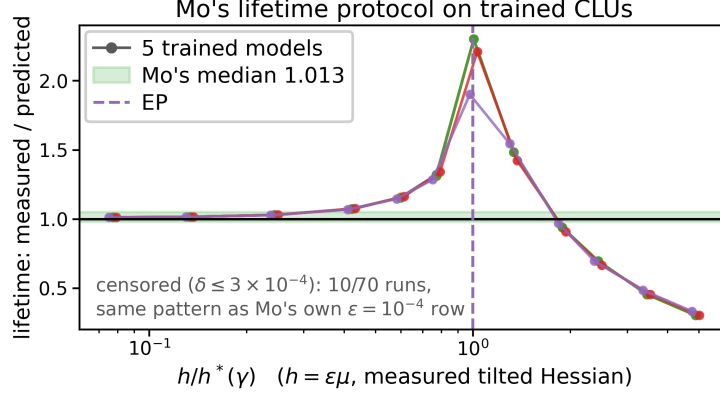


Figure 1: Mo’s single-exponential lifetime law, run *unchanged* on the trained models, is the over-damped face of the transverse-curvature price list. Measured/predicted lifetime ratio versus breaking magnitude, 14 regimes, 5 seeds: ≈ 1 overdamped, reproducing his reported median 1.013; a $2.2\times$ delay spike at the exceptional point; declining to $0.31\times$ deep underdamped — containment exact below the crossover, failing above it in the calculable ballistic direction. *Evidence*.

197 4.2 Evidence: a published lifetime law is the price list’s overdamped face

198 The price list should predict what other instruments see. *A published estimator gives the direct test,*
199 *because it was fitted to a different object. We therefore run it unchanged on our trained models*
200 *and read where it holds.* Mo (2026) shows that a “pseudo-gap” predicts a finite memory lifetime
201 through a single-exponential estimator. Running his exact code-level protocol — phase 0.35 rad,
202 threshold 0.2 rad, his censoring, cap 15000 steps — unchanged on the 5 trained models across
203 14 breaking magnitudes gives measured/predicted $1.012 \pm 0.000 \rightarrow 1.029 \pm 0.001$ overdamped
204 against his reported median 1.013, 2.202 ± 0.155 at the exceptional point, and 0.309 ± 0.012
205 deep underdamped. His law is the overdamped face of our price list: there the two coincide,
206 $\text{corr}(\log \text{pred}, \log \text{meas}) = 0.9987$ overdamped-only, and his predictor fails only past the crossover
207 — by $\approx 3.2\times$ at the deepest trained-model breaking we test, $\delta = 4$, our sweep going no deeper
208 underdamped — in the calculable ballistic direction, with the $2.2\times$ exceptional-point delay spike
209 as the third signature. On the *exact analytic map* the same mechanism continues to $\approx 5\times$ further
210 underdamped (theory note, check (k), $\gamma = 0.1$); the $\approx 3.2\times$ is the trained-model number, the $\approx 5\times$
211 is exact-map only, and we never quote it as evidence about a trained network. This is containment, not
212 conflict: below h^* we recover his reported number, and above it the regime structure — floor, ringing,
213 exceptional point — is what a first-order flow cannot exhibit and what a *constitutive* curvature adds.
214 It also does not rest on substituting the exact Jacobian gap for his predictor: the result reproduces on
215 Mo’s own finite-horizon estimator (Appendix E). *Evidence: a recently posted ML lifetime law, run*
216 *on trained checkpoints.*

217 4.3 The honest negative: where the price list does not extend

218 Symmetric training data might be expected to induce a flat direction on its own. *On this architecture*
219 *class it does not.* A generic MLP potential trained on that data develops *no* near-flat direction: softest
220 emergent $\mu^2 = 5.1/5.9/5.4 \times 10^{-2}$ on 3 seeds against the designed flat mode $\leq 2.4 \times 10^{-15}$, a gap
221 of 13–14 orders; Mo’s own attribution instrument separates the architectures by 15 orders. This is
222 a measurement on our architecture class and training recipe, not a general statement that learning
223 cannot produce a tuned flat direction — a local learning rule that does produce one is published
224 (Vafidis et al. 2022). Operationally that arm stores nothing on a *continuum*: ≈ 1 –1.6 bits on 2–3
225 washboard minima, a written δ relaxing completely, though the damping-optimum retention law itself
226 still generalizes to it. The continuous coset register is therefore designed-only, and we mark it as the
227 one face of the price list that does not transfer.

228 **The same boundary against learned baselines.** Gated and oscillatory recurrent networks fit this
229 task well. *Fitting it is not holding it. We therefore write a value into each and let it run autonomously.*
230 On Mo’s S^1 path-integration family the baselines are trained with a learning-rate sweep, best RMSE

231 kept, and are not strawmen: LSTM and LEM reach train-horizon RMSE 0.18/0.23 rad. Yet under
 232 Mo’s *autonomous* protocol the structural triad of latch, μ^{-2} law and bounded motion of the stored
 233 coordinate is qualitatively absent in them: coRNN/LEM/LSTM forget the stored analog phase in
 234 $\approx 5.6/56/69$ map-steps, the stored coordinate randomizes past 1.2 rad, and they are *perturbation-*
 235 *fragile*, a 0.1 hidden kick collapsing LSTM $69 \rightarrow 2$ and LEM $56 \rightarrow 5$, while the generically-trained
 236 unit holds ≈ 263 map-steps with bounded motion of that coordinate (≈ 0.35 rad, price-list law to
 237 $+12\text{--}15\%$) and the designed unit never moves (5/5 seeds). *Evidence.*

238 **Honest gap.** The load-bearing claim is the qualitative triad, architecture- and compute-independent.
 239 The raw 263-versus-69 map-step ratio does not survive compute normalization, and we retire it
 240 as a compute claim: per step the Verlet update costs $\approx 6.2\times$ an LSTM cell and $\approx 3.1\times$ a LEM
 241 cell in wall time, $\approx 14\text{--}15\times$ FLOPs, not width-matched, so per unit of retention the unit spends
 242 $\approx 23.5\times/\approx 14.6\times$ more wall time than the baselines it out-holds (Appendix C). Lacking native
 243 velocity ingestion, it cannot enter Mo’s input-driven task-RMSE axis at all, so no task-RMSE was
 244 fabricated.

245 4.4 The price list survives the training-time correction

246 A price list read off a potential is worth little if the training objective quietly rewrites the potential.
 247 *Ours does exactly that.* Wake–sleep contrastive divergence drives the order parameter r^* to zero,
 248 inverting the designed vacuum in 8/8 runs at the default 1000 epochs: the minimiser becomes
 249 invariant, with the degenerate μ^2 pair predicted in §3, on a horizon set by sleep-update *frequency*
 250 racing the wake clamp, and specific to flat or degenerate vacua. This is an instance of what the attractor
 251 literature calls the fine-tuning problem, the perturbation here being the training objective itself. *Scope*
 252 *clause, travelling with that finding:* our sweep varies chain length only over `sleep_steps` $\in \{50, 500\}$ at
 253 otherwise fixed sampler settings and does not resolve where either sits relative to this model’s mixing time, so
 254 the finding is *frequency-decisive across the two chain lengths we run*, and is not a claim that chain length is
 255 irrelevant to contrastive-divergence learning in general.

256 A $V(\text{data})$ energy anchor holds the condensate up — $\lambda = 100$: 5/5 seeds, $r^* = 0.911 \pm 0.016$, at
 257 the cost of weaker noise rejection and $\approx 35\times$ higher wake MSE. *That a corrective term can keep*
 258 *a flat direction alive is not new* (Renart, Song & Wang 2003). What we add is the third part of the
 259 price list: every headline law of it still holds under the correction, at $\approx 20\times$ the erosion horizon
 260 (Appendix A, Figure 2) — 3000 anchored epochs, 3 seeds, curvature ratio exact to 1.5×10^{-12} over
 261 4.6 decades, slope -0.956 , the same floor, and the exceptional-point onset bit-identical at 0.5165.
 262 The architecture *breaks* the symmetry, the objective *restores* it, the anchor *keeps it broken*.

263 5 Discussion

264 *Scope box — limitations and scope choices, stated so no reader has to infer them.* All results are laptop-CPU, dim
 265 4, the S^1 testbed, ≤ 5 seeds; the solvable core is quadratic and trained statements are local at a critical point of
 266 V_θ , with anharmonic deviations measured as a zero-kick retention bias of $+5$ to $+29\%$, kick-independent for 2/3
 267 emergent seeds and amplitude-dependent for the softest- μ^2 seed. The head-to-head is on *autonomous retention*
 268 in map-application units, and the unit does not enter the input-driven task-RMSE axis. BIBO boundedness is
 269 evidenced within the coercive-potential / compact-sublevel-set scope and concerns the autonomous, designed
 270 setting measured here — nothing here claims that symplectic structure buys off-distribution stability in a *trained*,
 271 *input-driven* setting, which we do not test and where a published symplectic architecture already holds the
 272 ground with a gradient bound — and friction never stabilizes a saddle (Appendix D). The symmetry framing is
 273 tree-level only. The finite- T diffusion law holds on both the designed channel (verification) and the emergent
 274 arm (evidence, 3 seeds, above $T^* \approx 3 \times 10^{-3}$) under FDT-consistent noise in a Newtonian kinetic mode; the
 275 one face that does not generalize is the continuous latch register, which is designed-only. (i) *Scale is out of scope*
 276 *by choice:* every number here is dim 4, hidden 64, ≤ 5 seeds, single-unit, laptop-CPU, and we report no result
 277 at larger dimension, wider layers, more units or accelerator scale. (ii) *No external benchmark is won on its own*
 278 *headline metric anywhere in this paper;* the comparisons are diagnostic retention protocols on a synthetic S^1
 279 family and matched-parameter ablations, not leaderboard results — and §4.3’s honest gap is part of the claim,
 280 not a caveat to it. (iii) *The substrate scope, once, in our own voice: these laws govern the store; end-to-end*
 281 *performance additionally depends on the encoder, measured separately, with its parameter and state-byte budget*
 282 *ledgered.*

283 A price list changes what a flat direction is for. Read this way a trained memory has latches — exact
 284 flat directions, the moduli of V_θ — and registers, nearly-flat modes of a chosen half-life, fenced as
 285 designed-geometry in Appendix D; kinetic isotropy is then the price of an equivariant write *current*,

not of the register. Named as directions, not results: the μ^{-2} law, crossover and floor re-measured at latent dimension $\gtrsim 64$ and $N \geq 8$ interacting units, where mode mixing can break the single-block reduction that makes the price list exact here; the head-to-head on a wider trained-model family; the anchored recipe at longer training and larger width; the equivariant-control wrapper that would put the unit on the supervised task axis; a learned-store re-test of the explicitly-broken cell respecting that fence; the map from (μ, γ, T) to lifetime whose $T = 0$ face is §4; and the price list re-derived beyond abelian $SO(2)$, for a torus T^2 or the non-abelian $SO(3) \rightarrow SO(2)$ coset — a direction, on which this paper offers no evidence.

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390 A The laws survive the training-time correction

391 B The price of the prior, the loan curve and the recovery ladder

392 *Demoted from the main text of this abstract, and retained in full: this appendix is a supporting result about what*
 393 *the physics prior costs, not one of this paper’s claims.*

394 **The price of the prior (evidence).** Against param-matched controls — dim 4, 150 epochs, $\pm 0.05\%$ params, 3
 395 seeds, same MLP potential — the picture is Pareto, not podium. Volume conservation buys bounded (BIBO)
 396 dynamics within the coercive-potential / compact-sublevel-set scope (1.00 versus 0.33 for a broken-volume
 397 arm), a protected near-flat direction (μ^2 0.008 versus 0.122) and vacuum survival under contrastive-divergence

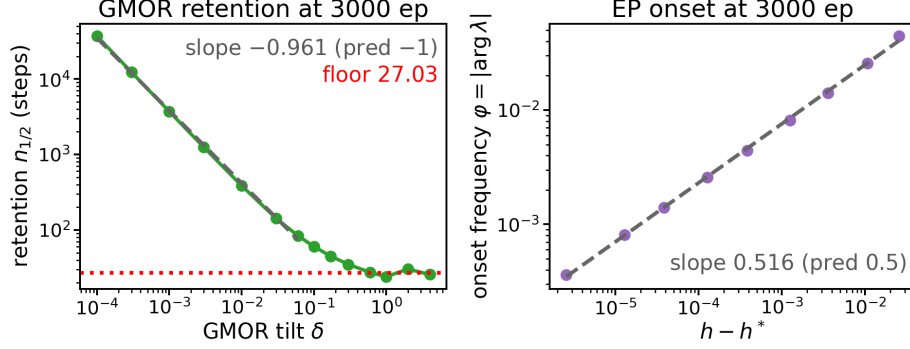


Figure 2: The headline laws of the price list survive the shipped anchor cure at 3000 epochs ($\approx 20\times$ the erosion horizon; anchored $\lambda=100$, 3 seeds). Left: the retention law $n_{1/2}$ versus tilt δ — anchored-3000-epoch points on the predicted slope, fitted -0.961 against predicted -1 , saturating at the curvature-independent floor 27.03; the μ^2/δ ratio is 1.0000 ± 10^{-12} over 4.6 decades. Right: the exceptional-point onset $\varphi \propto \sqrt{h - h^*}$ above the exceptional point, fitted slope 0.516 against predicted 0.5, with $\varphi = 0$ exactly below — bit-identical to the 150-epoch value. *Designed tested under the shipped cure; the survival of the laws through 3000 erosive epochs is evidence for the recipe.*

398 training ($r^* = 0.72$ versus 0); leapfrog buys the latch; and the prior costs raw fit, an unconstrained twin fitting
 399 $\approx 15\times$ better, 0.0128 against 0.190 eval MSE. That cost is *contraction-forbidden* by volume conservation —
 400 the broken-volume arm recovers $\approx 2.4\times$ of it precisely by leaking volume — and is not the causal velocity
 401 cap, inactive here. The loan is called at ≈ 700 steps (measured): the twin leads by 1–2 orders to ≈ 500 steps,
 402 crosses at ≈ 700 and diverges to 196 at 5000, while the unit holds a bounded plateau ≈ 0.20 – 0.23 . Its asset is
 403 boundedness by construction, not the lowest steady MSE — broken-volume (0.14) and LSTM (0.13) plateau
 404 below it — and of the licensed repairs only global damping buys the fit back, a learned friction *field* does
 405 not: paid mechanisms are not interchangeable. *The non-comparability caveat printed below travels with every*
 406 *number in this appendix.*

407 *Dim 4, hidden 64, learned-Newtonian kinetic term, dt = 0.05, circle $R=1\times 256$, 150 epochs, 3 seeds, wake-*
 408 *only objective except the γ_ϕ rung. Cumulative fit MSE against horizon on the stationary circle-vacuum orbit,*
 409 *param-matched to $\pm 0.2\%$ (CLU 4549 / broken-volume 4557 / twin 4551).*

horizon (steps)	twin	CLU	broken-volume	LSTM
10	2.9×10^{-6}	2.2×10^{-5}	1.1×10^{-4}	7.5×10^{-2}
50	6.2×10^{-5}	1.2×10^{-3}	1.0×10^{-3}	4.9×10^{-2}
100	2.4×10^{-4}	1.2×10^{-2}	4.1×10^{-3}	4.9×10^{-2}
500	1.1×10^{-2}	1.8×10^{-1}	9.7×10^{-2}	7.7×10^{-2}
1000	3.2×10^{-1}	2.0×10^{-1}	1.2×10^{-1}	9.9×10^{-2}
5000	1.96×10^2	2.2×10^{-1}	1.4×10^{-1}	1.3×10^{-1}

411 The twin is best by 1–2 orders to ≈ 500 steps, crosses above CLU at ≈ 700 and diverges to 196 at 5000; CLU is
 412 the only arm flat *by construction*, on a bounded plateau ≈ 0.20 – 0.23 . Broken-volume (0.14) and LSTM (0.13)
 413 plateau slightly *below* CLU (0.22): “physics costs raw fit” persists as a small steady-state penalty, and CLU’s
 414 asset is boundedness, not the lowest plateau.

415 ***Non-comparability caveat, to be read before using these numbers.*** This ladder is a **separate wake-only**
 416 **experiment**, and its **absolute MSE scale and its twin (0.0047) are not cross-comparable** with the param-
 417 **matched table above** (twin 0.0128, different seeds, different training objective). **Only within-table gaps are**
 418 **meaningful**. In particular, do not divide across tables: the “ $\approx 15\times$ ” above (0.190/0.0128) and this table’s
 419 implicit “ $\approx 44\times$ ” (0.2066/0.0047) are the *same qualitative statement* measured on two different scales, not
 420 two different findings, and the 92% below is a fraction of **this** table’s absolute gap (0.202) only. The $\gamma = 0$
 421 conservative rung (0.2066) is this experiment’s counterpart of the $\gamma = 0$ CLU of the param-matched table
 422 (0.190).

rung	eval MSE	% gap recovered	BIBO	latch drift	flat μ^2
twin (no physics)	0.0047	100% (def.)	1.00	—	—
CLU conservative ($\gamma = 0$)	0.2066	0% (def.)	1.00	0.778	5.2×10^{-3}
+γ global ($\gamma = 0.05$)	0.0216	92%	1.00	0.778	5.2×10^{-3}
+γ_ϕ learned field ($K = 2$)	0.2548	−24%	1.00	0.706	2.1×10^{-2}

Each licensed mechanism is added one at a time; the gap recovered is $(\text{CLU}_{\gamma=0} - \text{rung}) / (\text{CLU}_{\gamma=0} - \text{twin})$, twin gap 0.202. Global γ recovers 92% of the **absolute** gap at zero structural cost, leaving BIBO, latch and μ^2 unchanged, while still trailing this table’s twin by $\approx 4.6\times$ in *ratio* (0.0216 against 0.0047); both statements must travel with the claim. The learned γ_ϕ field is a targeted-forgetting tool and does not recover fit, at -24% ; that rung retrained with the sleep phase on, so the “wrong tool” verdict is robust but the -24% is not a clean single-knob delta.

C Per-step compute: CLU-Verlet against LSTM and LEM

*Single-core CPU, f32, XLA; this backs the retirement of the “ $\approx 4\times$ longer” as a compute claim in §4.3. One CLU dissipative-Verlet step (kick–drift–kick, two $\nabla_q V$ backprops through a $\text{dim-4} \rightarrow 64 \rightarrow 64 \rightarrow 1$ tanh MLP, learned-Newtonian kinetic term, $\gamma=0.05$, $\text{dt}=0.05$) is measured against one LSTM and one LEM cell, both input 1, hidden 16, output 2, the exact retention baselines of §4.3. Wall time is the median of 7 scan-amortized repetitions over 2×10^5 steps; the naïve single-call timing is dispatch-bound at a $\approx 5 \mu\text{s}$ floor and is *not* reported.*

unit	params	FLOPs/step	wall ns/step (scan-amortized)
CLU Verlet (h64, dim 4, KDK)	4549	36148	1554
LSTM cell (h16)	1186	2512	252
LEM cell (h16)	1186	2400	500

normalization	CLU vs. LSTM	CLU vs. LEM
raw map-steps (263 vs. 69/56)	$3.8\times$ longer	$4.7\times$ longer
FLOP-normalized	$54.8\times$ more	$70.7\times$ more
wall-normalized	$23.5\times$ more	$14.6\times$ more

Per-step ratios: CLU/LSTM = $14.4\times$ FLOPs and $6.2\times$ wall; CLU/LEM = $15.1\times$ FLOPs and $3.1\times$ wall. The $4\times$ **inverts** under compute normalization: per unit of retention the unit spends $\approx 15\text{--}24\times$ the wall time and $\approx 55\text{--}71\times$ the FLOPs of the RNN it out-holds. **Confound, flagged:** the comparison is not width-matched, CLU at hidden 64 with two gradient evaluations against baselines at hidden 16; a width-matched CLU would narrow the gap, but the sign is robust because the Verlet step requires two potential backprops, and the 263/69/56 retention numbers come from these specific configurations. The measurement is single-core CPU at batch 1; GPU or batched throughput could shift the wall ratios, though not the FLOPs. §4.3 accordingly leads with the qualitative triad, states the per-step $\approx 6\times$ (LSTM) and $\approx 3\times$ (LEM) wall factor, and does not carry the map-step ratio as a compute claim.

D Prominent negatives

Negatives are recorded, never dropped. Each row states the claim tested, the result, and the number that settles it.

claim tested	result	number
Does a learned inertial mass isotropize on symmetric data?	No; and the breaking is invisible to the Hessian and to the latch, showing only as a bounded Noether-charge oscillation.	Log-mass split ratios 0.98 / 1.18 / 5.4 from init to 150 ep; charge amplitudes 5.4/1.6/0.082 $\times 10^{-2}$; tied controls exactly 1.0000.
Does wake-sleep contrastive divergence preserve a designed degenerate vacuum?	No — it inverts it. The cure is a $V(\text{data})$ energy anchor, and the headline laws survive it (§4.4).	$r^* \rightarrow 0$ in 8/8 runs at 1000 ep; inversion at epoch 116/442/959 by sleep frequency; anchor $\lambda=100$ holds 5/5 seeds at $r^*=0.911 \pm 0.016$.
Does an emergent (MLP) unit have a continuous coset register?	No — the continuous register is designed-only; an emergent unit stores a few bits on discrete washboard minima (evidence, 3 seeds; §4.3).	$1 - \lambda_{\text{coset}} \approx 10^{-3}$ against the designed $\leq 1.1 \times 10^{-15}$, ~ 12 orders; written δ retained $\leq 2.1 \times 10^{-3}$; capacity $\approx 1\text{--}1.6$ bits.
450 Do raw $n_{1/2}$ exponents discriminate designed from emergent?	No — instrument, not physics: a matched designed control on the same grid gives the same slopes, both carrying the ℓ_θ/Δ boundary layer. No $n_{1/2}$ may be quoted without its Δ and ℓ_θ/Δ .	Designed control $d \log n_{1/2}/d \log T = -0.53, -0.60, -1.04$ and $d \log n_{1/2}/d \log \gamma = +0.78, +0.63, +0.55$; ℓ_θ/Δ up to 2.0.
Is an explicit-breaking coefficient a lifetime dial on a learned store?	No, and the first half fails in sign: raising the breaking <i>lowers</i> the softest eigenvalue, and a coercivity term, not the breaking, sets the ceiling. A designed degeneracy does not survive superposition.	$\lambda_{\min}$ runs $+0.0994 \rightarrow -8.2846$ and $+0.3291 \rightarrow -1.1980$ over four decades of breaking; hard ceiling $\tau_{\max} = \Gamma/2\alpha = 4.0$; tilt-vacuum residual 0.140–0.343 against a random-direction baseline $1/\text{dim} = 0.167$; tangential curvature predicted 0.100, measured 0.0994.
Can this unit enter the input-driven path-integration task-RMSE axis?	No — it has no native velocity ingestion and the equivariant-control wrapper is unbuilt. No task-RMSE was fabricated.	—
claim tested	result	number
Do a learned friction field and a curvature governor compose usefully?	No — the pair is Pareto-dominated by either alone, and a learned compact-support gate under-covers without accurate placement.	Locus discovery 2/6 before adaptive- K , 2/3 after; best learned rejection 0.861; a λ re-sweep does not approach the oracle.
Can friction stabilize a saddle, or an energy gate catch isoenergetic escape?	No to both — a genuine scope limitation of the class. Only curvature control of V_θ and the mass-anisotropic cap $v_i^{\max} = c/\sqrt{M_i}$ bound expanding or isoenergetic modes.	$\lambda_+ > 1$ for all γ , with $\min_\gamma(\lambda_+ - 1) = 1.4 \times 10^{-5}$.
451 Is the coded momentum-coupled Langevin sampler FDT-exact in every kinetic mode?	No — exact only in the Newtonian modes; in the relativistic mode <i>no</i> noise scale gives the coded chain a Gibbs invariant at all.	Required per-mode scale $\sigma_i^* = \sqrt{M_i T \gamma (2 - \gamma)}$.
Is a mean-spectrum chaos regularizer informative?	No — it is degenerate; a chaos penalty must use a max or positive-part statistic.	—
Is <code>sleep_temperature</code> active at <code>sleep_friction=0</code> ?	No — a silent knob; cite it when reproducing temperature-swept sleep runs.	$\sigma \propto \sqrt{\gamma_{\text{sleep}}}$, so the swept runs are bit-identical.

452 **A reporting caution travels with the isotropization row.** The closed-form charge-oscillation amplitude is an
453 **on-vacuum-orbit** statement; **off the orbit only boundedness holds**. The amplitude-to-split ratio is moreover
454 **not** constant, as the exact envelope requires, since it is linear only to leading order — so no proportionality
455 constant should be quoted as a law, and we report amplitudes and boundedness, never a “drift rate”. The design
456 rule attached to mass-tying is correspondingly about the *current*, not the register: **an anisotropic channel still**
457 **latches with infinite half-life**; tie the masses for an angle-independent write current, and at $\dim(G/\mathcal{H}) \geq 2$ a
458 degenerate pseudo-Goldstone multiplet, **not** to save the register.

459 **Two reading rules travel with the learned-store row.** (a) The breaking coefficient of that battery is an
460 **explicit-symmetry-breaking magnitude**; it is **not** this paper’s integrator step ε , and the two must never be
461 conflated. (b) No sentence in this paper may be read as claiming that a tilt magnitude is a lifetime dial on a
462 learned store, or that a pseudo-Goldstone lifetime can be extended without bound by shrinking the breaking: on a
463 store with a coercive confinement $\alpha\|q\|^2$ the ceiling is $\Gamma/2\alpha$, and lowering α breaks the write. This paper’s own
464 tilt results (Appendix F) are analytic tilts on **designed** architecturally-invariant potentials, where the construction
465 is exact; the learned-store scope is exactly what this negative removes. That battery is a *different architecture*
466 from the trained units of this paper and shares none of their training configuration, which is why it is quoted
467 as a scope fence and never as a measurement of the units here. What survives it is the *geometry*: the identity
468 $\lambda_{\min} = \epsilon$ holds in the single-atom geometry in which it was derived, unit-tested to 5%, and the registered
469 damped-mode floor $\tau = 2/\Gamma$ is confirmed, 5.0 predicted against 4.23–6.58 measured time units.

470 **The sampler row’s scope, stated precisely, because the qualifier is load-bearing.** The coded damping substep
471 is an additive Gaussian momentum kick of fixed covariance, so the chain’s invariant momentum marginal is
472 necessarily Gaussian-smoothed; for a *relativistic* kinetic term the Gibbs momentum marginal is Maxwell–Jüttner,
473 which is not of that form (*theory note*, Prop-9’ — a characteristic-function argument needing neither $V = 0$ nor
474 linear damping). **The failure is in the sampler, not in the thermodynamics**: the configurational equilibrium

475 $\pi_q \propto e^{-V_\theta/T}$ is relativity-insensitive, since the momentum integral factorizes out, so a relativistic unit’s
 476 equilibrium distribution is perfectly well-defined and the chain simply fails to sample it; a one-line thermostat
 477 carrying a randomized, momentum-dependent inertia equal to the relativistic mass repairs it. We therefore never
 478 assert that a relativistic unit “has no equilibrium”. **This touches no result in this paper:** the trained units here
 479 are `newtonian_learned` throughout, so σ^* is exact for every configuration we run — the qualifier is a *scope*
 480 *clause* on a class-level theorem, not a correction to a measurement.

481 **E Positioning: the four retirements in full, and the head-to-head on Mo’s own** 482 **estimator**

483 *The four retirements of §2, elaborated so that every number behind them is on the record.*

484 **(1) Attention and state-space models have retention-guarantee analogues.** HiPPO-LegS (Gu et al. 2020)
 485 has an $O(tL/\sqrt{N})$ approximation bound, a polynomial $\Theta(1/t)$ rather than exponential gradient decay, and
 486 *exact* timescale equivariance with no discretization step to tune — exact where ours is a discrete map at fixed ε
 487 with $h = \varepsilon\mu < 2$ a hard stability limit — and we have no measurement saying a learned curvature spectrum is
 488 preferable to it. Tunable permanence is likewise occupied (EDEN, Karuvally et al. 2025); UnICORN (Rusch
 489 & Mishra 2021b) is Hamiltonian, structure-preserving and *has* the gradient-bound theorem we lack; and the
 490 modern Hopfield update *coincides with* attention for a single head with identity projections and $\beta = 1/\sqrt{D}$
 491 (Ramsauer et al. 2021).

492 **(2) “Retrieval is a rollout, not a weighted sum” is not our novelty.** The move is Kong, Brewer & Lai’s (2024),
 493 at capacity $N_c \propto K^{1.08 \pm 0.01}$ for their one-hot and binary index codings and 1.17 ± 0.02 for their 2D coding.
 494 Our differences are structural: neither symplectic, Hamiltonian nor mass-structured there, and the address enters
 495 through an index channel rather than as the initial condition of the state.

496 **(3) Continuity is not an escape hatch from the NTM/DNC lineage.** Both were already fully soft (Graves et al.
 497 2014, 2016), so the precedent transfers as a warning. Of the three DNC failure modes Csordás & Schmidhuber
 498 (2019) diagnose, key/value entanglement, which they rank first, is architectural in a memory whose address q_0 is
 499 both the key and the initial condition of the value; de-allocation aliasing maps onto any position-gated forgetting;
 500 chained-read degradation is the third. None are measured here.

501 **(4) Not a competitor to the transformer.** Never claimed, and indefensible at this scale: ≈ 1 –1.6 bits per
 502 emergent register, no win over a trivial baseline on any external benchmark, and formal limits on fixed-size
 503 latent state for copying and retrieval (Jelassi et al. 2024) that conservation does not evade — it prevents *decay*,
 504 not *capacity*.

505 **The head-to-head reproduced on Mo’s own instrument.** §4.2’s result does not depend on substituting the
 506 exact Jacobian gap for Mo’s predictor. With Mo’s own finite-horizon estimator $\hat{\lambda}(T=128)$ the result holds on
 507 his instrument: $\text{corr} = 0.9995$ overdamped, $\text{meas/pred } 0.86$ – 1.03 , and 0.30 at $\delta = 4$ against the exact-gap
 508 0.31 .

509 **F The price list verified in full, and GMOR proper on trained checkpoints**

510 *Designed testbed with analytic tilts and an analytic spurion, therefore verification of the theory’s exactness;*
 511 *5 seeds at 150 epochs plus 3 anchored at 3000 epochs, dim 4, laptop CPU, f64 probes, probe-only with no*
 512 *retraining. The retention law $n_{1/2} \propto \mu^{-2}$ is a Gell-Mann–Oakes–Renner (GMOR) law in the spectral mass;*
 513 *we use the name for the identity $\mu^2 F^2 = \delta\Sigma$ throughout this appendix. **GMOR proper — §F.3 onward — is***
 514 *demoted from the main text of this abstract and retained here in full; it is a supporting result, not one of this*
 515 *paper’s claims.*

516 **F.1 The price list, verified on trained models — the battery behind §4.1.** §4.1 states this battery’s headline
 517 quantities: the constitutive identity to 3.2×10^{-10} over all 70 tilt rows on a machine-flat trained vacuum
 518 ($\mu^2 \leq 2.4 \times 10^{-15}$); the GMOR law $\mu_{\text{meas}}^2/\mu_{\text{pred}}^2 = 1.000000 \pm 5 \times 10^{-12}$ at every δ and seed across 4.5
 519 decades; the fitted overdamped retention slope -0.985 ($n = 35$; predicted -1); the curvature-independent
 520 floor 27.03 steps at $\gamma = 0.05$; the metric bifurcation (0.01% agreement overdamped, a $3.2\times$ split at $\delta = 4$);
 521 the frequency onset ($\varphi = 0$ exactly below h^* on 15/15 rows, fitted slope 0.5165 against $1/2$ above); and the
 522 latch transport law to $\leq 1\%$ over two decades of γ with the written angle frozen to $\leq 1.2 \times 10^{-15}$ rad. Four
 523 measurement details belong here rather than there. The $\sqrt{h - h^*}$ onset holds down to $h - h^* = 2.6 \times 10^{-6}$
 524 on a fine multiplicative grid ($h - h^* \in [-4.2 \times 10^{-3}, 2.6 \times 10^{-2}]$); the slight slope excess over $1/2$ is far-field
 525 bending. The onset prefactor is 0.268 against the theory note’s 2×2 -normalized 0.3247, a one-line units
 526 reconciliation. Where the quality factor permits ($f = 2, 4$) the trajectory frequency matches the Jacobian
 527 frequency to 0.06–0.3%; near onset the mode decays before one period completes, so the exceptional point is
 528 spectroscopically real but dynamically silent there. And the eigenvalue pair below h^* is real, which is what
 529 makes $\varphi = 0$ an exact statement rather than a resolution limit.

F.2 The damping corollary — a supporting result, deliberately not a headline. The same block fixes the half-life’s dependence on the damping, and that dependence is **not** monotone: sweeping γ at fixed spectral mass, $n_{1/2}(\gamma)$ traces a V-shape of depth $\approx 8\times$ whose argmin sits at, or just below, the exact critical-damping value $\gamma^* \approx 2\varepsilon\mu$, on all 5 designed seeds. Below γ^* the mode is underdamped at the floor $2\ln 2/(-\ln(1-\gamma))$, which *shortens* as γ grows; above it the mode is overdamped at $n_{1/2} \approx 2\gamma \ln 2/[(2-\gamma)(\varepsilon\mu)^2]$, which *lengthens*; the exceptional point is where the branches meet. Two consequences: (i) a retention measurement quoted at one γ is uninformative about the sign of $\partial n_{1/2}/\partial\gamma$ unless the band is named, the same architecture giving both signs; (ii) the $\mu \rightarrow 0$ latch corner is the limit $\gamma^* \rightarrow 0$, so a Goldstone register is permanently overdamped, which is the structural reason friction cannot erase it.

F.3 GMOR proper: why a second spurion. The angular tilt $V \rightarrow V + \delta(1 - \cos n\theta)$ of F.1 is *radius-independent* — measured on these checkpoints, $\max |r_{\text{tilt}}^* - r^*| = 2.22 \times 10^{-15}$ over all (checkpoint, δ) pairs — so it measures only the **product** $\mu^2 F^2 = \delta n^2$, in which the condensate degenerates to the pure number n^2 (measured: $\max |\mu^2 F^2/(\delta n^2) - 1| = 1.03 \times 10^{-11}$); the correct instrument for verifying a power law, and provably incapable of resolving the condensate. The **linear ambient spurion** $V \rightarrow V - \delta(u \cdot q)$ — the exact analogue of the ChPT quark-mass term, with u in the channel plane — lets the vacuum radius run with δ , so the three GMOR objects become separately measurable on a trained checkpoint: the pseudo-Goldstone spectral mass μ^2 from the autodiff Hessian at the polished vacuum; the decay constant $F^2 = M_{\text{ch}} r^{*2}$, the learned channel inertia times the vacuum radius squared and **not** the orbit radius; and the condensate $\Sigma = r^*(\delta) = -\partial E_{\text{vac}}/\partial\delta$. Over 8 trained checkpoints and $\delta \in [10^{-8}, 0.3]$ (10 values, 8 decades), $\max |\mu^2 F^2 - \delta\Sigma| = 1.33 \times 10^{-15}$ over all 80 pairs.

δ	$\Sigma = r^*$ (mean)	$\mu^2 F^2$ (mean)	max rel. dev.	max abs. dev.
10^{-8}	0.95186937111604	$9.5186937286 \times 10^{-9}$	1.10×10^{-8}	1.01×10^{-16}
10^{-7}	0.95186948174437	$9.5186948223 \times 10^{-8}$	1.87×10^{-9}	1.85×10^{-16}
10^{-6}	0.95187058802597	$9.5187058802 \times 10^{-7}$	1.87×10^{-10}	1.86×10^{-16}
10^{-5}	0.95188165067351	$9.5188165068 \times 10^{-6}$	2.18×10^{-11}	2.16×10^{-16}
10^{-4}	0.95199226031843	$9.5199226032 \times 10^{-5}$	1.34×10^{-12}	1.33×10^{-16}
10^{-3}	0.95309668422658	$9.5309668423 \times 10^{-4}$	2.27×10^{-13}	2.09×10^{-16}
10^{-2}	0.96398369127653	$9.6398369128 \times 10^{-3}$	2.69×10^{-14}	2.69×10^{-16}
3×10^{-2}	0.98732513581811	$2.9619754075 \times 10^{-2}$	9.66×10^{-15}	2.71×10^{-16}
10^{-1}	1.06435252346294	$1.0643525235 \times 10^{-1}$	1.55×10^{-15}	1.53×10^{-16}
3×10^{-1}	1.60757655208590	$4.8227296563 \times 10^{-1}$	1.55×10^{-15}	1.33×10^{-15}

Precision fine print, next to the claim. GMOR holds to machine precision in **absolute terms at every** δ ; relative exactness ($\leq 2.7 \times 10^{-14}$) is demonstrated for $\delta \geq 10^{-2}$. The growing *relative* deviation at small δ is **not** a failure of the law but the roundoff floor of the **autodiff Hessian**: the angular curvature is $K_{\theta\theta} = \delta/r^*$, reconstructed as a difference of $O(\|K\|)$ terms, so its absolute error is $\sim \epsilon\|K\|$ and its relative error $\sim \epsilon/\delta$. This was confirmed directly: on the analytic Mexican hat a **cancellation-free** Hessian returns relative deviation $1.1\text{--}2.2 \times 10^{-16}$ at every $\delta \in [10^{-8}, 0.3]$, while the autodiff Hessian on the same potential returns $2.28 \times 10^{-8} \rightarrow 4.22 \times 10^{-15}$ across that range, exactly $\propto 1/\delta$. **Do not quote “ 2.2×10^{-16} relative” for this experiment.** The same ϵ/δ floor applies to every μ^2 probe in this paper; all other sweeps here use $\delta \geq 10^{-4}$, where the floor is $\sim 10^{-12}$ and the quoted relative agreements are sound.

F.4 Σ is genuinely measured, three ways; it runs, and the angular tilt cannot see it. $\Sigma_{\text{geom}} = r^*(\delta)$; $\Sigma_{\text{HF}} = u \cdot q^*(\delta)$ by Hellmann–Feynman, agreeing to $\max |\Sigma_{\text{HF}} - \Sigma_{\text{geom}}| = 0.0$ exactly; and $\Sigma_{\text{FD}} = -dE_{\text{vac}}/d\delta$ by central finite difference at $h=10^{-6}$ — the envelope-theorem definition, structurally independent of the Hessian — agreeing to $\max |\Sigma_{\text{FD}}/\Sigma_{\text{geom}} - 1| = 9.2 \times 10^{-11}$. Under the linear spurion Σ runs by +16.1% to +210.1% across the δ grid (anchored-3000-epoch models +19.8/ +17.4/ +16.1%; designed-150-epoch models +210.1/ +169.3/ +35.9/ +34.4/ +41.1%), while under the angular tilt on the *same* checkpoints it does not move at all ($\leq 2.22 \times 10^{-15}$). With u at channel angles $\{0.0, 0.7, -2.3\}$ on one designed checkpoint at $\delta=10^{-2}$, the μ^2 relative spread is 6.21×10^{-15} , as exact channel invariance requires.

F.5 The leading low-energy constant is saturated by the radial resonance. With $\mu_{\text{LO}}^2 \equiv \delta\Sigma(0)/F^2(0)$, the theory predicts a *relative* leading-order error $x \equiv \delta/(M_{\text{ch}}\mu_{\text{rad}}^2 f)$. Measured: the ratio is 0.99999607, the mean over 8 checkpoints at $\delta \leq 10^{-5}$, maximum deviation 2.20×10^{-5} ; from the measured μ^2 over $\delta \in [10^{-4}, 10^{-2}]$ it lies in $[0.97917, 0.99992]$; and corrections are first order, $d(\text{ratio})/dx = -1.0504$ at small x , i.e. ratio = $1 - O(x)$ — exactly what ChPT demands of a *leading* low-energy constant. The CLU therefore realizes ChPT’s **resonance saturation of low-energy constants** exactly rather than phenomenologically, the radial (σ) mode being the saturating resonance. **Breakdown is visible and expected:** at $\delta = 0.3$ the softest-radial seed, $\mu_{\text{rad}}^2 = 0.670$, has $x = 0.68$ — not small — and Σ has run +210%; the expansion is uncontrolled there, so $\delta = 0.3$ **must not be quoted for the next-to-leading-order claim**: cap at $x < 0.25$ or state x explicitly.

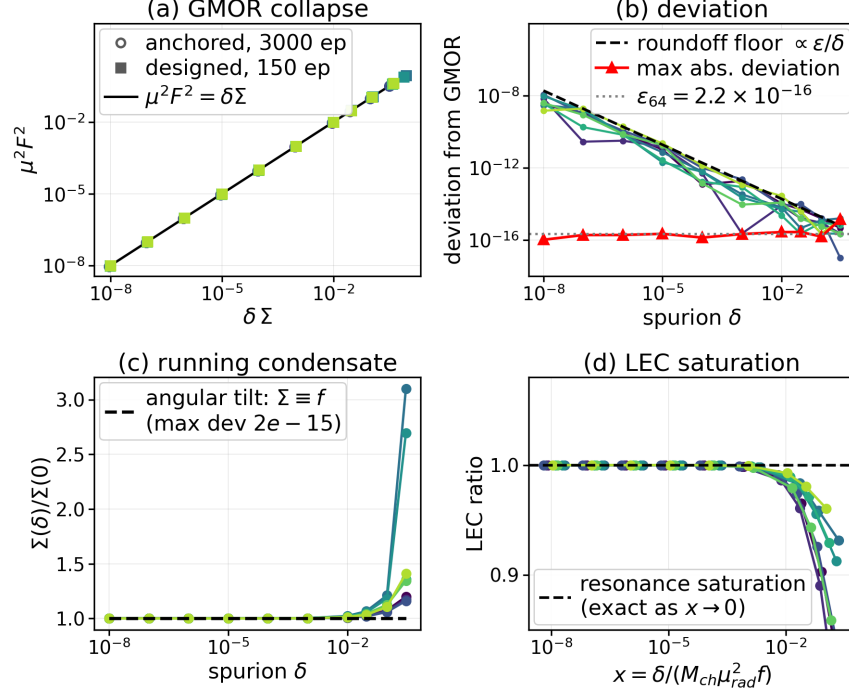


Figure 3: GMOR proper on trained checkpoints, with a linear ambient spurion. (a) The collapse $\mu^2 F^2 = \delta \Sigma$ over 8 decades of δ , 8 trained $SO(2)$ checkpoints. (b) Absolute deviation pinned at the f_{64} floor ($\leq 1.33 \times 10^{-15}$) against the ϵ/δ relative-roundoff line of the autodiff Hessian — the small- δ relative residual is a probe artifact, not a law violation. (c) The running condensate $\Sigma(\delta)/\Sigma(0)$ under the linear spurion against the angular tilt’s exactly flat $\Sigma \equiv r^*$ — the shipped tilt cannot see the condensate. (d) The low-energy-constant ratio tending to 1 as the chiral expansion parameter $x \rightarrow 0$, with slope -1.05 . *Designed testbed, analytic spurion, trained checkpoints, therefore verification.*

578 **F.6 Honest scope.** This appendix is tree-level and classical: no loops, chiral logarithms, running low-energy
579 constants or anomalies; “resonance saturation” is an exact algebraic statement about a two-mode classical
580 potential, not a phenomenological fit. It is probe-only on already-trained checkpoints, spurion off by default so
581 as to be bit-compatible with every result elsewhere, on one architecture family — designed exact $SO(2)$, dim 4,
582 laptop-CPU. The emergent MLP arm is untested and would carry its own self-breaking δ_{eff} , so GMOR there
583 should be tested as $\delta \rightarrow \delta + \delta_{\text{eff}}$: an open falsifiable, not a result.